

Ryan Ye

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Education

Cornell University

Ithaca, NY

B.S. in Computer Science, College of Engineering, GPA 4.02

Expected May 2028

- **Coursework:** Deep Learning, Computer Vision, Machine Learning, Systems Programming (C), Algorithms, Data Structures, Functional Programming (OCaml), Discrete Math, Linear Algebra, Differential Equations, Multivariable Calculus
- **Honors:** BURE Fellow, CIDA Grant Recipient, AIMI Junior Affiliate, 4-time AIME Qualifier, USACO Silver Division

Technical Skills

- **Languages:** Python, Java, C/C++, OCaml, JavaScript (Node.js, Express.js, React), HTML/CSS
- **Libraries:** PyTorch, JAX, Hugging Face Transformers and PEFT, NumPy, pandas, Matplotlib
- **Tools:** Git, SSH, SLURM/HPC, CUDA, Jupyter, VS Code

Research Experience

Undergraduate Researcher, Sun Lab at Cornell University

May 2025 – Present

PI: Jennifer Sun. AI for science: scientific discovery agents and computer vision for animal behavior

- **Statistical validation for automated hypothesis discovery (AIMI Junior Affiliate project), Summer 2026.** Found and fixed an admission-conditioning bug that let an MCTS-based discovery system's false discovery rate run 35–41% against a 10% target; shipped two corrected registration modes
- Showed that reusing holdout data under adaptive tree selection pushed the false discovery rate to 99.3% at scale, which led to the conservative controller becoming the default; built the validation layer from scratch (anytime-valid evidence interface, e-value fixes, interval-null tests) and benchmarked 8 controllers on four real-world datasets
- **Multi-lab video behavior recognition (MABe), Feb 2026 – Present.** Beat the frozen-backbone baseline on 9 of 15 lab datasets (13 of 15 on at least one split) by LoRA fine-tuning a frozen video foundation model with a self-authored kinematic auxiliary loss; on one lab, video alone beat the pose-tracking ceiling
- Built the 8-stage SLURM pipeline with 5-fold cross-validation and ensemble threshold calibration across all 15 datasets, staging up to 1.9 TB of video per lab on node-local NVMe; brought it up on Blackwell GPUs (CUDA 12.8)
- **Automated calf behavior analysis (BURE Fellow, CIDA grant), May 2025 – Present.** Raised best posture-classification F1 from 0.928 to 0.957 across 24 runs comparing DINOv2 and DINOv3 ensembles on 300 to 1,000 images; fine-tuned YOLO on calf data I annotated in CVAT. Poster presented at ADSA 2026; manuscript in preparation

Projects

Worship Rig

Sep 2026

Personal project. Electron, JavaScript, Web Audio

Site · GitHub

- Built and shipped a Mac app that turns a MIDI keyboard into a keys rig for worship bands: a Web Audio engine with 43 instruments and a key drone, a GarageBand sample converter, and 1,000+ automated tests

DoRA Re-Implementation and Evaluation

Spring 2026

CS 4782 Deep Learning final project, 4-person team. PyTorch, Hugging Face PEFT

GitHub

- Team re-implementation of DoRA, tested against LoRA on Qwen2.5-3B (AMC12, MATH); ran all HPC experiments
- Built the evaluation harness (logit-probe and two-pass scoring separating what a model knows from what it generates), the data pipeline with a leakage filter, and the analysis and figures; diagnosed a critical bug in the DoRA layer

Traveling Salesman: Conversational Flight Search

Mar 2026

Cornell AI Hackathon, frontend. React, d3-geo

GitHub

- Built most of the React frontend: the trip planner, ranked flight plans, and routes on a 2D map and 3D globe

Sisyphus: AI Productivity App

Nov 2025

Claude Builders Club Hackathon, 4-person team. Node.js, Express.js, Claude API

GitHub

- One of 5 winners of 38 projects (Fun Award): checks photo check-ins with Claude; worked on the frontend and AI validation

Publications

- M. M. Ferreira, K. M. Dadallage, T. Ward, H. McCray, **R. M. Ye**, J. J. Sun, F. A. Leal Yepes. “A smarter lung ultrasonography solution for lung consolidation classification in dairy calves using a deep learning approach.” *Journal of Dairy Science*, 2026. Supporting author
- **R. M. Ye**, M. Madureira Ferreira, J. J. Sun, F. A. Leal Yepes. “Automating pose detection in calves: a computer vision framework for non-invasive behavioral monitoring.” Poster, ADSA Annual Meeting, 2026
- **R. Ye**, Y. Chen, K. A. Kelly. “The effects of firm performance on CEO compensation and CEO pay ratio before and during COVID-19.” *Research in Economics* 77(4), 453–458, 2023. First author

Leadership

Basic Coding, Co-Founder. Free coding classes for 300+ students in 8 states during COVID-19

Aug 2020 – Jun 2024